

LESSON 6-7 • SESSION 1 • PRACTICE • FORM B

Factors and Greatest Common Factor

Learning target – I can list the factors of two numbers and find their greatest common factor. · 6.NS.B.4

REMEMBER

- A **factor** divides a number with no remainder. A **common factor** divides both numbers.
- The **GCF** is the greatest factor two numbers share. Check it by dividing it into both.
- **Prime factorization**: split until every factor is prime:
 $63 = 7 \times 3 \times 3$.

WORKED EXAMPLE

Find the GCF of 20 and 30.

Step 1 Factors of 20: 1, 2, 4, 5, 10, 20.**Step 2** Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30.**Step 3** Shared factors: 1, 2, 5, 10. The greatest is 10.**Answer GCF = 10****WORDS** **factor** (*factor*) · **greatest common factor** (*máximo común divisor*) · **prime factorization** (*factorización prima*)**PRACTICE***show your work · label units*

1. Find the GCF of each pair.

a. 24 and 42 _____**b.** 33 and 55 _____**c.** 20 and 28 _____**d.** 14 and 35 _____

2. **Factor Pairs.** Work with the number 63.

Hint Start with 1×63 , then try 2, 3, 4, 5, ...**a.** List every factor pair of 63. _____**b.** List all factors of 63 from least to greatest. _____

3. **Prime Factor Tree.** Build a factor tree for 70 in the box.

Hint Split 70 into 7×10 , then split 10.**a.** Prime factorization of 70: _____**b.** Is 5 a factor of 70? _____*Show your work.*

4. **Pet Packs.** A shelter worker has 20 chew toys and 35 treats. Every pack must hold the same number of each, with nothing left over.

a. Greatest number of packs: _____**b.** In one pack: chew toys ____ treats ____*Show your work.*

5. **Ivan's Answer.** Ivan says the GCF of 18 and 24 is 3. Is he right? Give the correct GCF.

Hint Is there a shared factor greater than 3?

Show your work.

GCF _____

6. **Mei's Answer.** Mei says the GCF of 22 and 55 is 2. Is she right? Give the correct GCF.

Hint List the factors of 55.

Show your work.

GCF _____

7. **Garden Rows.** A gardener has 36 carrot seeds and 45 bean seeds. She plants identical rows with none left over.

- a. Greatest number of rows: _____
b. In one row: carrot seeds ____ bean seeds ____

Show your work.

8. **Party Bags.** A host has 40 stickers and 56 erasers for identical bags with none left over.

- a. Find the greatest number of bags. _____
b. How many of each item go in one bag? _____
c. Explain why 10 bags would not work. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Devon's Work.** Devon wrote: " $18 \div 9 = 2$, so the GCF of 18 and 42 is 9." Explain his mistake. Then find the correct GCF.

TEST PRACTICE

choose the best answer

10. The GCF of 16 and 20 is –

A 2 B 8 C 80 D 4

11. Which number is a common factor of 12 and 20?

A 3 B 4 C 6 D 10

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

2 × 5 × 7 D	8 E	9 D	3 Q	5 I	6 V	11 I	4 X	2 Z
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#3a	#4a	#5	#6	#7a	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____